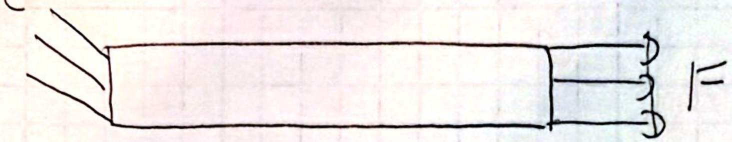
Outline



I. Selected Values

Load = 400 lbf

Max deflection axial = 0.009 in

Aluminum Bar

Young Modulus = 10×10^6 psi

Diameter = 0.500 in

$$A = \frac{\pi d^2}{4} = \frac{\pi (0.500)^2}{4} = 0.19635 \text{ in}^2$$

$$\delta = \frac{FL}{AE}$$

$$L = \frac{\delta A E}{F} = \frac{(0.009)(0.19635)(10{,}000{,}000)}{400}$$

$$= 44.18 \text{ in}$$

Stress check

$$\sigma = \frac{F}{A} = \frac{400}{0.19635} = 2037\,psi = 2.04\,ksi$$

$S_y = 40\,ksi$

$$n = \frac{S_y}{\sigma} = \frac{40}{2.037} = 19.6$$

Axial Deflectron

$\delta calc = 0.009 in$

$\delta FEA = 9.016 \times 10^{-3} in = 0.009016 in$

$$\% = \frac{|\delta FEA - \delta calc|}{\delta calc} \times 100$$

$$= \frac{|0.009016 - 0.009|}{0.009} \times 100 = \boxed{0.178\%}$$

$\delta max = 0.009 in$

$\delta FEA = 0.009016 in > \delta max$

$\sigma_{vmax} = 2.233 ksi$

$$\sigma = \frac{F}{A} = \frac{400}{0.19635} = 2.037 ksi$$

$$\frac{2.233 - 2.037}{2.037} \times 100 = \boxed{9.62\%}$$

$S_y = 40 ksi$

$$n = \frac{S_y}{\sigma_{VM}} = \frac{40}{2.233} = \boxed{17.91}$$